School of Engineering

REPUBLIC POLYTECHNIC

AY2025 SEMESTER 2 – GRADED ASSIGNMENT 1 (GA1) for CA1

Module Code : E211

Module Name : Operations Planning II

Item	Submission Date	Upload softcopy of assignment to:
Report (Individual)	30 November 2025 11:59:00 pm	SA3.0 File Submission
Presentation Slides (Team)	5 December 2025 11:59:00 pm	POLITEMall

INSTRUCTIONS TO STUDENTS

1. E211 Graded Assignment 1 (GA1) consists of the following:

   a) GA1 Report (Individual) – 75%

   b) GA1 Presentation (Team) – 25%

2. Answer all questions.

   a) Prepare your Report (Individual) in a single MS Word Document with a minimum of 1000 words.

   b) Prepare your Presentation Slides (Team) in a single PPT of no more than 20 slides.

3. Name your Report (Individual) clearly, using ***E211_GA1_Your Student ID***. Include a cover page in your document, indicating your name and student ID. **Zip the report together with other files such as FlexSim models and Excel (if any) into one ZIP FOLDER.**

4. Name your Presentation Slides (Team) clearly, using ***E211_GA1_Your Team Number***. Include a cover page in your PPT, indicating name and student ID for each member in your team.

5. You should allocate sufficient time to complete and submit the assignment according to the deadlines given above. Late submissions without Leave of Absence (LOA) will be subjected to the following late penalty:

Time after submission deadline	Between 0 and ≤ 24 hours	Between 24 and ≤ 48 hours	Between 48 and ≤ 72 hours	After 72 hours

Percentage of total marks deducted by	10%	20%	30%	100%

6. Do note that plagiarism of work and/or cheating will lead to failure of your Graded Assignment and possibly more severe penalties which include disciplinary action, suspension or even dismissal from the polytechnic.

   Provide a proper list of references at the end of your report.

   Include the following declaration in your GA submission:

   "I, <Student Name (Student ID)>, declare that the assignment submitted is not plagiarized and is entirely my own work, and that no part of it has been copied from any work produced by other person(s) or source(s). I am aware that issuing a false declaration can result in severe penalties. I understand that I will be subjected to disciplinary actions if I am found to have committed any act of academic misconduct. I am aware that acts of academic misconduct include (but are not limited to) the following: submitting work that is not entirely my own work, engaging in unauthorized collaboration on the assignment that is meant to be done individually, facilitating academic dishonesty by allowing another student to submit my work (in part or as a whole) as his/ her work."

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Individual Report (75%)

You are to demonstrate your ability to study a real-life queuing system using simulation. In this assignment, your aim is to model a queuing system in FlexSim simulation, analyze results of the simulation and study a what-if scenario to assess the impact to the system performance. Answer all parts and provide your response in a report. You are to submit your report together with FlexSim models and any other files in ONE ZIP FOLDER.

Part 1 (10 marks)

Create a queuing scenario for your study by inputting the following prompt in ChatGPT or similar AI tools.

"Generate a simple scenario for a 2-stage XX queuing system. There should be multi-servers in at least one of the stages. Include 30 sets of randomly generated data for the inter-arrival times and service times."

Select one of the words below to replace the 'XX' in your prompt.

- Logistics
- Manufacturing
- Healthcare
- F&B
- Retail

Inter-arrival times - LESSON 5 (Set Source Parameters)
- Arrival Style: Inter-arrival times

Service times - L5 Processor time in Processors (Set Processor Parameters)
In FlexSim, Process Time is the duration the processor spends working on an item.

a. Describe the scenario in your own words by elaborating on the AI generated text. State any assumptions, inherent issues and performance measures of the system you are studying. (8 marks)

(maximum 3)
Waiting time, waiting length, utilization ..

b. Insert a **screenshot of your ChatGPT prompt and response** here. Also include the corresponding ChatGPT link**.** (2 marks)

Part 2 (28 marks)

a. Build a FlexSim simulation model to simulate the queuing system you described in Part 1. (18 marks)

Your model should contain the following:

- Appropriate objects that are properly named and connected to simulate the process
- Correct data inputs based on generated arrival and service times
- Dashboard charts/ tables displaying the relevant performance measures
- Experimenter set up and run to generate confidence interval results of the performance measures

b. Provide these details in your report: (7 marks)

why u use separator, this is represent what?

- Briefly explain the type of 3D objects used in your model
- Briefly explain how the distributions for arrival and service times are determined
- State the results of the performance measures from simulation

c. Identify and justify any bottleneck in the queuing system. (3 marks)

Part 3 (12 marks)

what if this happen? etc.

a. Craft a realistic what-if scenario for the queuing system you described in Part 1. (5 marks)

b. Modify your FlexSim model to simulate the what-if scenario and generate the results. (3 marks)

c. Compare and analyze the performance measures between the original scenario and the what-if scenario. Finally, provide an appropriate conclusion for your study. (4 marks)

Team Presentation (25%)

Instructions

Each team is required to submit a deck of GA1 Presentation Slides and carry out an in-class presentation.

1. Your communication skills (CS) will be assessed individually through the presentation.

2. The CS grading is based on the CS Rubrics given in SA3.0.

3. Each team member is to prepare a few slides to reflect on what they learnt from building the FlexSim simulation model.

4. Each team will be given 20 minutes for the presentation.